

Cybersecurity

ML and AI in Cybersecurity Defense

Lesson 2.2



Learning Objectives

By the end of today's class, you will:

- 1 Understand how AI is used in cybersecurity defense.
- 2 Learn about specific AI/ML techniques for threat detection and prevention.
- 3 Gain hands-on experience with AI tools used in cybersecurity defense.



AI and ML Techniques for **Cybersecurity Defense**

AI and ML Techniques for Cybersecurity Defense

AI is a crucial tool in protecting systems from cyber threats.

AI and ML techniques significantly enhance our ability to detect and respond to cyber attacks in real-time.



Automation and efficiency



Real-time analysis



Managing data volumes



AI and ML Techniques for Threat Detection

Three primary methods:

Anomaly detection

Identifies deviations from normality. Crucial for spotting potential security breaches that don't match any known patterns.



Signature-based detection

Signature-based detection uses known patterns of malicious activity to detect threats.



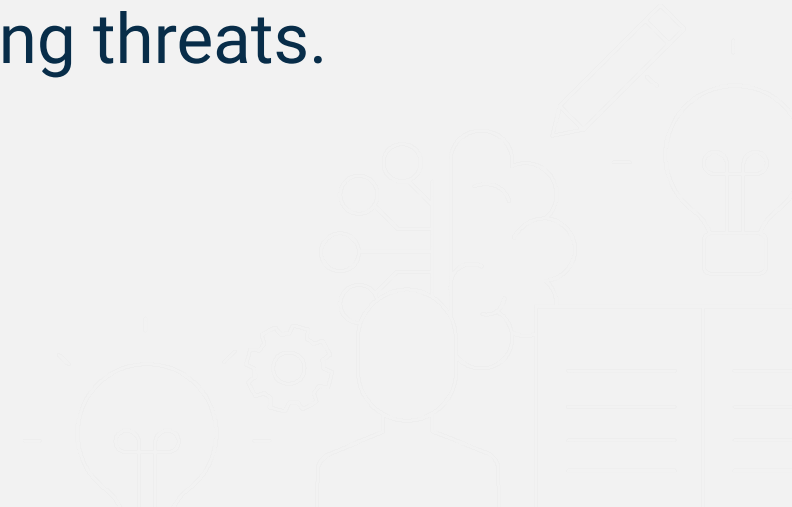
Behavioral analysis

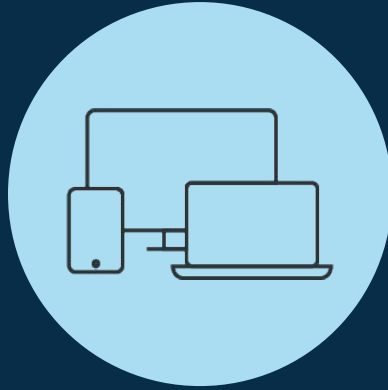
Behavioral analysis monitors user behavior for suspicious activities.





AI and ML also plays a crucial role in preventing threats.

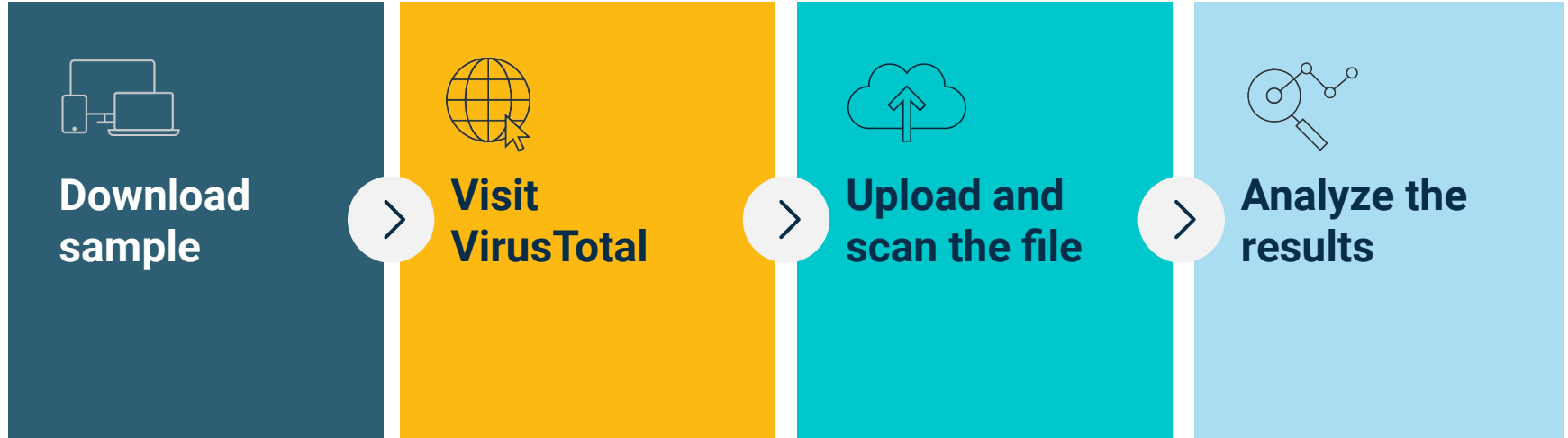




Instructor Demonstration

AI and ML in Threat Prevention

AI and ML in Threat Prevention





Activity:

Identifying Virus Threats

In this activity, you will use AI to analyze several viruses and how to mitigate their effects.

Suggested Time:
15 Minutes





Time's up!
Let's review

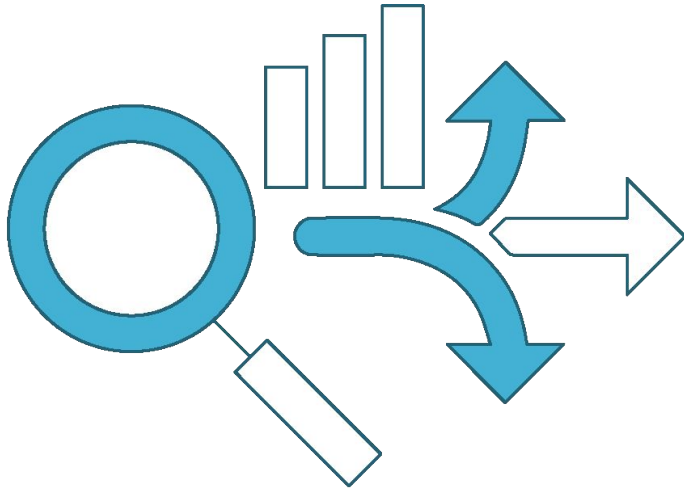


Questions?



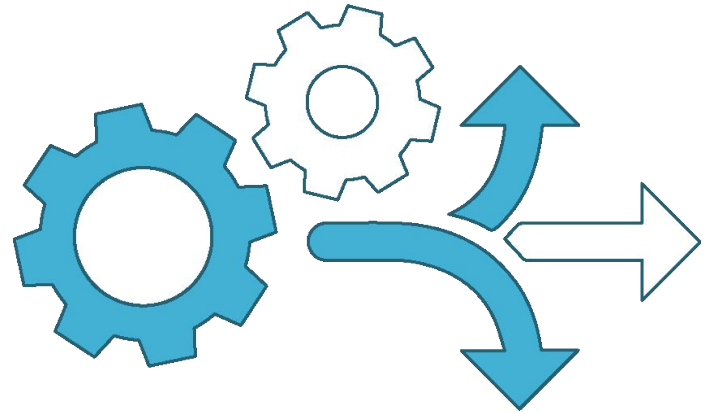
Predictive Analytics

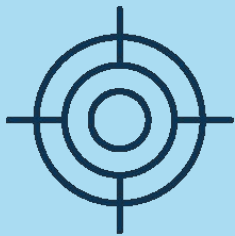
Analyzing historical data to
forecast future threats



Automated Responses

Automatically responding to
detected threats

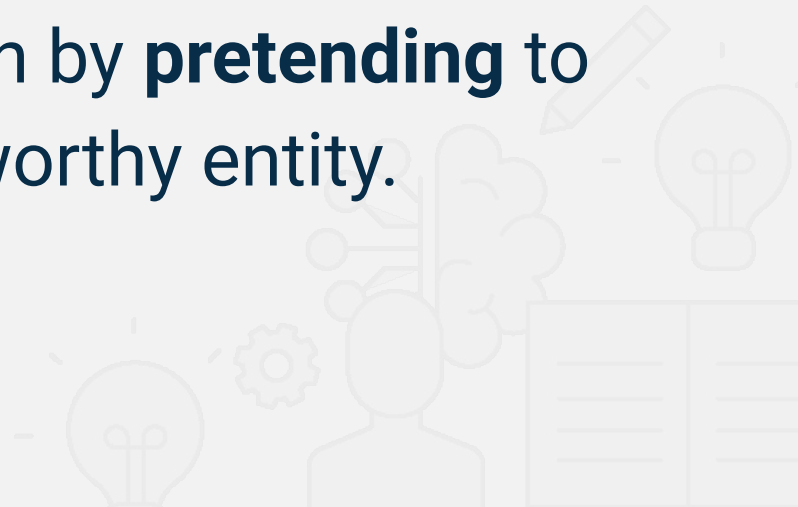




AI and ML to Prevent **Phishing Attacks**

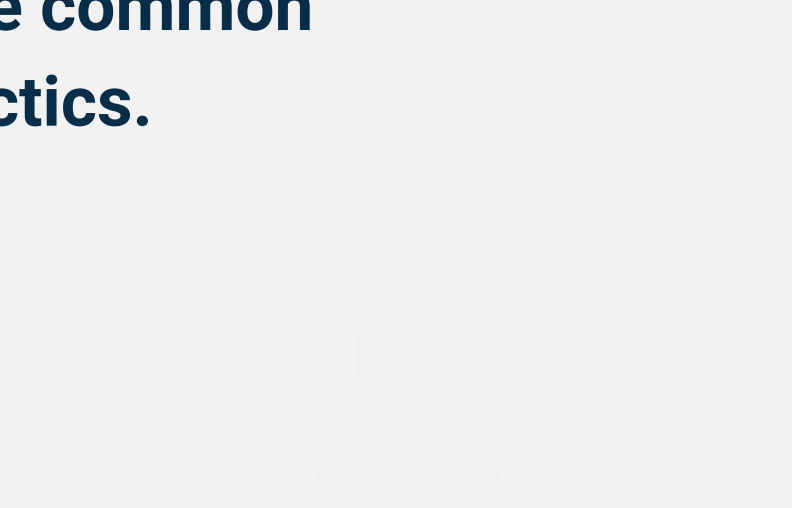


Phishing involves **fraudulent attempts** to obtain sensitive information by **pretending** to be a trustworthy entity.





AI models can be trained
to **recognize common
phishing tactics.**





Natural Language Processing

Analyzes text content of emails and messages

Detecting phishing attempts by identifying unusual phrasing, grammatical errors, or urgency cues



SUBJECT: URGENT!

Dear: Customer

We requires you to
verify your account...





URL Analysis

Examines URLs for signs of phishing

Identifying fraudulent websites designed to mimic legitimate ones, such as slight misspellings of popular domain names.



<https://microsoft.com.office>

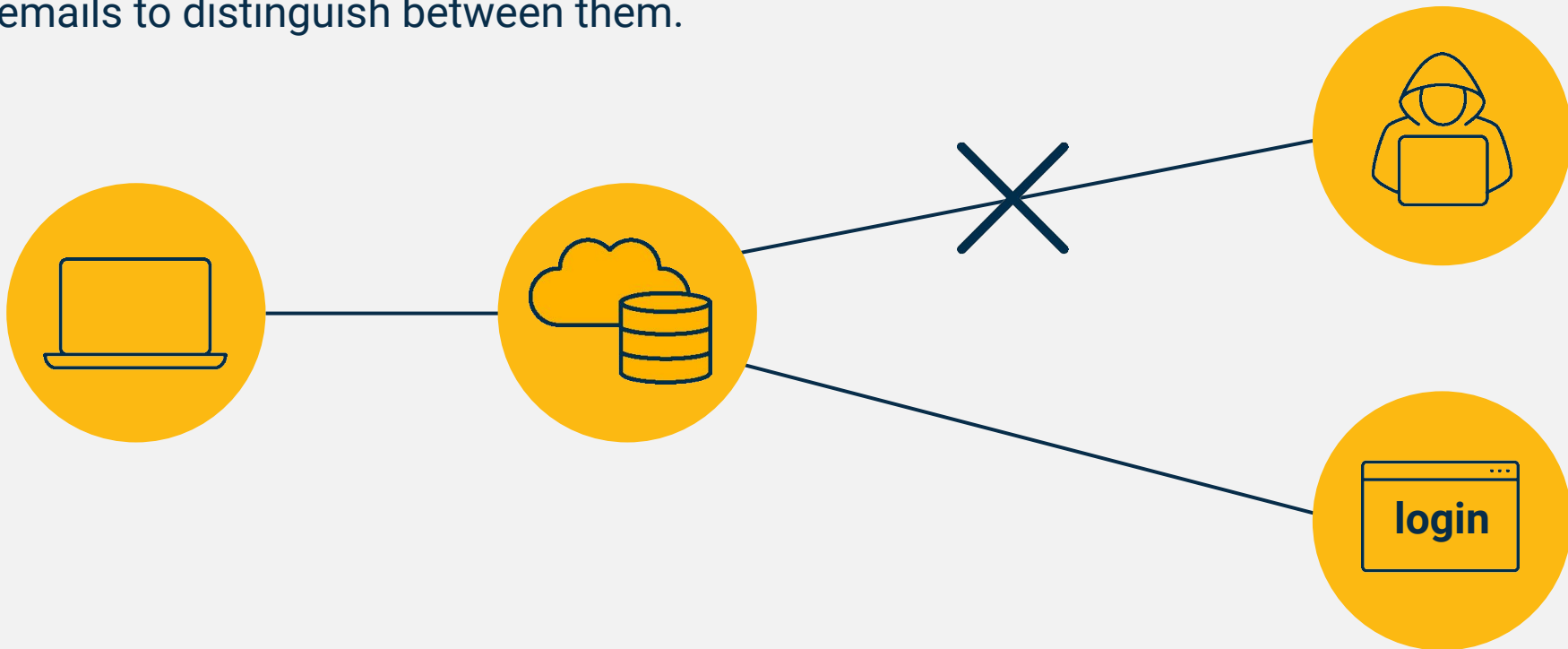


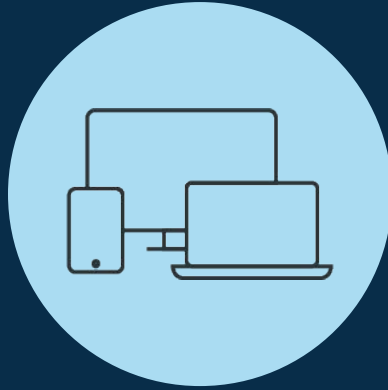


Machine Learning Classification

Uses supervised learning models to classify emails and websites

Training a classifier on a dataset of known phishing and legitimate emails to distinguish between them.





Instructor Demonstration

Identifying Spam with AI



Dear Valued User,
Congratulations!

You have been selected randomly to win the
GRAND PRIZE of **\$1,000,000** from the Global
Lottery Commission.

Click the link below to claim your prize immediately.
This offer is only valid for the next **24 hours**.

Claim your prize now

Best regards,
Global Lottery Team



This email example is a classic representation of a phishing attempt, often flagged as spam by email services like Gmail. Here's why it would likely be marked as suspicious:

1. **Unsolicited Claim of a Prize:** The email starts with a grand announcement of winning a significant amount of money, which is a common tactic in scam emails to grab the recipient's attention.
2. **Sense of Urgency:** By stating the offer is only valid for the next 24 hours, it creates a sense of urgency, pushing the recipient to act quickly without due diligence.
3. **Suspicious Link:** The text "Claim Your Prize Now" is linked to a potentially malicious or deceptive site (indicated here by `#`). Such links are often used to steal personal information or install malware.
4. **Generic Salutation:** The use of a generic greeting like "Dear Valued User" instead of personalized information suggests a lack of legitimacy, as the sender does not know the recipient personally.
5. **Lack of Detailed Contact Information:** Legitimate emails from real organizations usually include detailed contact information, not just a generic signature.

These characteristics make the email highly suspicious, and modern email filters are adept at recognizing such patterns to protect users from potential scams.



Activity:

Identifying Spam with AI and ML

In this activity, you will be analyzing several emails to see if AI will categorize them as legitimate or malicious.

Suggested Time:
15 Minutes



Time's up!
Let's review

Identifying Email Threats with AI

Common themes ChatGPT discovered

01

Analysis personalization

02

Content and offer

03

Call to action

04

Urgency

05

Sender information





Questions?





Break
15 mins



AI and ML Techniques for **Malware Detection**



Malware comes in various forms, each with its unique characteristics and methods of attack.

Types of Malware



Viruses:

Malicious code that attaches itself to files and spreads within a computer



Worms:

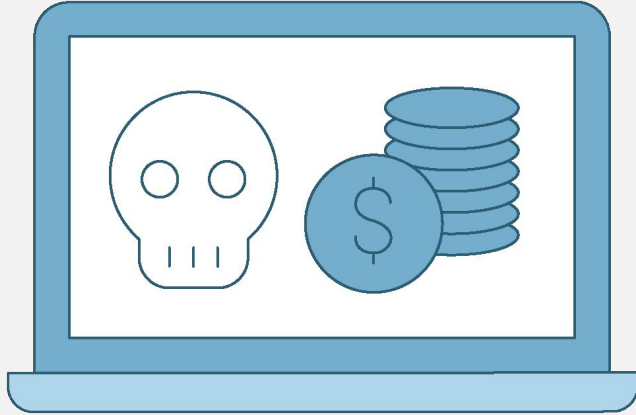
Malware that replicates itself to spread to other computers



Trojans:

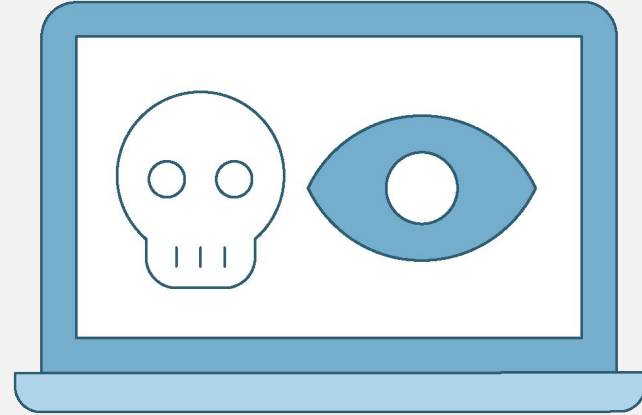
Malicious software disguised as legitimate software

Types of Malware



Ransomware:

Malware that encrypts the victim's data and demands a ransom



Spyware:

Software that secretly observes user's activities and reports it to the software's author

Using AI and ML to Detect and Analyze Malware

01

Data analysis

- AI and ML can analyze large volumes of data to identify patterns and signatures indicative of malware.



02

Pattern recognition

- AI models can be trained to recognize common malware characteristics and behaviors. This allows them to detect even new and unknown malware types based on their actions rather than relying solely on known signatures.



Antivirus Software



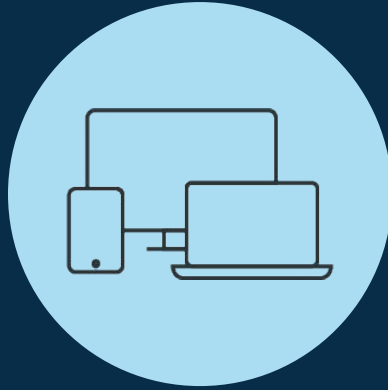
Tools like Windows Defender and Norton use AI and ML to detect and analyze malware.

- Continuously updated databases
- Uses advanced algorithms to identify and mitigate threats
- Can recognize known and new malware

Security Information and Event Management (SIEM) Systems

- Integrate AI and ML for real-time malware detection and response
- Collect and analyze security data from various sources
- Enable rapid neutralization
- Provide real-time monitoring and response capabilities





Instructor Demonstration

Identifying Malware



ChatGPT: “Summarize how the Wineloder malware works and its effects.”



Winloader Malware: Overview and Impact

How Winloader Malware Works

Winloader is a sophisticated backdoor malware used in cyber espionage campaigns. Here's a breakdown of its operation:

1. Initial Infection:

- The malware is often delivered through phishing campaigns, typically disguised as legitimate documents or invitations. In the case of the SPIKEDWINE campaign, it was disguised as an invitation from the Ambassador of India ([Check Point Research](#)).

2. Execution:

- Once the victim opens the malicious document, the embedded code executes and installs Winloader on the system. This malware leverages advanced techniques to avoid detection and establish a foothold on the infected device.

3. Persistence:

- Winloader is designed to maintain persistence on the infected machine, ensuring it can survive reboots and continue to operate covertly. This involves modifying system settings and creating scheduled tasks.

4. Command and Control (C2):

- The malware connects to a comma ↓ and control server to receive instructions from the



ChatGPT: “How do you mitigate
this malware?”



Mitigating Wineloder malware involves a combination of preventative measures, detection strategies, and response actions. Here are key steps to mitigate this malware:

Preventative Measures

1. Employee Training and Awareness:

- Educate employees about the risks of phishing attacks and train them to recognize suspicious emails and links. Regular training can significantly reduce the likelihood of falling victim to phishing schemes used to deliver Wineloder.

2. Email Security:

- Implement advanced email filtering solutions that can detect and block phishing emails and malicious attachments before they reach users. Use email authentication methods like SPF, DKIM, and DMARC to prevent spoofing.

3. Endpoint Protection:

- Deploy robust endpoint protection solutions that include antivirus, anti-malware, and intrusion detection systems. These tools can identify and block malicious activities and files on devices.

4. Regular Software Updates:





Activity:

Identifying Malware with AI and ML

In this activity, you will be researching common malware programs and how to mitigate the damage.

Suggested Time:
20 Minutes





Time's up!
Let's review



Questions?





Let's recap



Recap

ML and AI in Cybersecurity Defense

01

AI/ML play a critical role in detecting and preventing malware.

02

There are different malware types.

03

AI/ML can analyze data to identify malware patterns and behaviors.

04

AI/ML is used in antivirus software and SIEMs.



Questions?





Thank you